



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df049-2018-apr-android
Operator:	Floris
Analysis Started:	2026-06-19 08:40:52
Analysis Completed:	2026-06-19 08:42:31
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df049\2018_april\android
Output Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df049\2018_april\dfdof_output\ft-df049-2018-apr-android
Evidence Sources:	2
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
android_logical.zip	controller_android	logical	715.4 MB	2026-06-19 08:40:54
df049_internal.E01	drone_sd	physical	7.8 GB	2026-06-19 08:41:18

Evidence Source Path

C:\Users\Floris\Documents\ddj_drones\train_mavic_air\df049\2018_april\android

Cryptographic Hashes

android_logical.zip		controller_android	logical
SHA-256:	9fbc9082643497ed0a8dad1e04fc6d1b60bc1646825fef7d243c15f9fdf363a0		
SHA-1:	4012a8e785b8c6752a63bff302fc8dac9dbe0075		

df049_internal.E01		drone_sd	physical
SHA-256:	ce48d2b0d1b634442795a80c804053ef25bbaa6013e97eacc166458331a4a8d8		
SHA-1:	4663934cb481d453bd7afe709c81069fff5adde		

Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p1 - drone sd]: mmls: no partition table found in df049_internal.E01; treating as direct partition image
2. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for android_logical.zip
3. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for android_logical.zip
4. [p2 - controller ios]: source evidence not found
5. [p3 - controller android]: installed DJI apps not found in P2 output and no known DJI app directories found in archive, extraction aborted
6. [p3 - controller ios]: source evidence not found
7. [p3 - drone flight storage]: source evidence not found
8. [p4 - controller ios]: source evidence not found
9. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
exiftool	10	10	[OK]
fls	1	1	[OK]
icat	10	10	[OK]
mmls	1	0	[X] Failed
parser_android	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	2 / 7
Flights With Primary Correlation	0 / 0
Tools Succeeded	4 / 5

3. Device and Account Identification

No device or account information could be extracted.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	Yes
controller_ios	No
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
drone_sd	0	0	0	0	0	6	4

Data Quality Notes per Category

Images (6 files):

- 6 file(s) missing (normalisable) EXIF timestamp
- 6 file(s) missing EXIF GPS coordinates

Videos (4 files):

- 4 file(s) missing (normalisable) EXIF timestamp
- 4 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

No data quality issues recorded.

Pipeline Anomaly Flags

1. [p1 - drone sd]: mmls: no partition table found in df049_internal.E01; treating as direct partition image
2. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for android_logical.zip
3. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for android_logical.zip
4. [p2 - controller ios]: source evidence not found
5. [p3 - controller android]: installed DJI apps not found in P2 output and no known DJI app directories found in archive, extraction aborted
6. [p3 - controller ios]: source evidence not found
7. [p3 - drone flight storage]: source evidence not found
8. [p4 - controller ios]: source evidence not found
9. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Images

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0006.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos

DJI_0001.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0006.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

- exif_contains_no_gps: is/are missing EXIF GPS coordinates

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no installed DJI applications found - Device and account information.
2. Note: no controller device name found - Device and account information.
3. Note: no drone serial number found - Device and account information.
4. Note: no drone model found - Device and account information.
5. Note: no controller device found - Device and account information.
6. Note: no dji application version found - Device and account information.
7. Note: no account e-mail found - Device and account information.
8. Note: no product type found - Device and account information.
9. Note: no firmware version found - Device and account information.
10. Note: no last country code found - Device and account information.
11. Note: no aircraft model code found - Device and account information.
12. Note: no controller serial number found - Device and account information.
13. Note: no last known latitude found - Device and account information.
14. Note: no last known longitude found - Device and account information.
15. Note: no account uid found - Device and account information.
16. Note: no account nickname found - Device and account information.
17. Note: no device uuid found - Device and account information.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

DJI_0001.THM	Images	drone_sd	extract_physical
SHA-256:	7b7d8a5f04de4720da9c1d65924375c0a19425649b715c9be9d95f431ffe48ef		
SHA-1:	4b066c18aff86e0949ee2429e960119a060b5ad4		

DJI_0002.JPG	Images	drone_sd	extract_physical
SHA-256:	a859aac8fa7aef573e8ae8ae3880d8f3ee02b1325af52634cb8409c1afa94303		
SHA-1:	7b1610a6d6348d5fa470b9d560ca5233305c4cec		

DJI_0003.JPG	Images	drone_sd	extract_physical
SHA-256:	d1dcc50fa8edd7d2c5533bfa132ab0ce83711bcd731b2618ac375493204c98db		
SHA-1:	0eb3c492ab8c8ede7b38c21cf79a61735e6ec7f5		

DJI_0004.THM	Images	drone_sd	extract_physical
SHA-256:	da822f8e97c7453374509b92551f91fc5823f61c969d8f372600bf034ffa3feb		
SHA-1:	0b4527095c4e76844df4dc7c3a6177486c2fa103		

DJI_0005.THM	Images	drone_sd	extract_physical
SHA-256:	ba8631c371d71a3ff5a502566ebdf3772e25a1b88794300f943035795106b0e7		
SHA-1:	d33a50de0b692f83b445759004924cb4cf0e636e		

DJI_0006.THM	Images	drone_sd	extract_physical
SHA-256:	45f1f4e8ca1a8e9a319a7c16ac93ff020646e480121e85cf9768363713777e41		
SHA-1:	8ab6ad224bc3360a44ac2380500a15435abccbb8		

DJI_0001.MP4	Videos	drone_sd	extract_physical
SHA-256:	1457d7112e401e547986c4f0c172f896353d62e4b5d973c949b5353fc1c55778		
SHA-1:	95ca55621f71eb8e63873546809c13e97d753715		

DJI_0004.MP4	Videos	drone_sd	extract_physical
SHA-256:	efab6df387144c83cf98e56c8c83bf15db89d4d810f8b9afc7f57df2e004c095		
SHA-1:	5b51e7202edd35a65ce852c88fc5da0fe2248b31		

DJI_0005.MP4	Videos	drone_sd	extract_physical
SHA-256:	5a82b3a1d0bab3d575fd3a0884bd34012b0744368e672bf9bf568793021c44cf		
SHA-1:	032c213f233635ae0e9407b448d5fc80cbd70e80		

DJI_0006.MP4	Videos	drone_sd	extract_physical
SHA-256:	fdda7dbce2012b8258cde0832897d72373d0882a311936fdd4772c0af14edb83		
SHA-1:	68d706e06c209a9717a152cbc5a9eef5f615eb17		

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	6	6	4
Videos	4	4	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-19 08:40:52 mmls vThe Sleuth Kit ver 4.15.0 rc=1

cmd: mmls.exe df049_internal.E01

2026-06-19 08:40:52 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p df049_internal.E01

2026-06-19 08:41:31 parser_android v- rc=n/a

cmd: android_logical.zip C:\Users\Floris\Documents\dfdof_output\ft-df049-2018-apr-android\p2_image_parsing\controller_android_parsed
out: controller_android_parsed

2026-06-19 08:41:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1357829

out: DJI_0001.MP4

2026-06-19 08:41:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1357830

out: DJI_0002.JPG

2026-06-19 08:41:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1357832

out: DJI_0003.JPG

2026-06-19 08:41:45 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1357834

out: DJI_0004.MP4

2026-06-19 08:42:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1357835

out: DJI_0005.MP4

2026-06-19 08:42:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1357836

out: DJI_0006.MP4

2026-06-19 08:42:27 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1358853

out: DJI_0001.THM

2026-06-19 08:42:27 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1358854

out: DJI_0004.THM

2026-06-19 08:42:27 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1358855

out: DJI_0005.THM

2026-06-19 08:42:27 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df049_internal.E01 1358856

out: DJI_0006.THM

2026-06-19 08:42:28 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM

out: DJI_0001_exif.json

2026-06-19 08:42:29 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.JPG

out: DJI_0002_exif.json

2026-06-19 08:42:29 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.JPG

out: DJI_0003_exif.json

2026-06-19 08:42:29 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.THM

out: DJI_0004_exif.json

2026-06-19 08:42:29 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.THM

out: DJI_0005_exif.json

2026-06-19 08:42:30 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0006.THM

out: DJI_0006_exif.json

2026-06-19 08:42:30 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MP4

out: DJI_0001_exif.json

2026-06-19 08:42:31 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.MP4

out: DJI_0004_exif.json

2026-06-19 08:42:31 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MP4

out: DJI_0005_exif.json

2026-06-19 08:42:31 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0006.MP4

out: DJI_0006_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.